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Multivariate count data regression models and their applications

Ayman Alzaatreh¹, Felix Famoye², Carl Lee³

Abstract Multivariate regression models based on multivariate discrete distributions will be defined and studied. Multivariate discrete distributions, including some distributions generated from the $T-R\{W\}$ method will be defined. Models that allow both positive and negative correlation between any pair of response variables will be considered. The model parameters will be estimated by using the method of maximum likelihood estimation. The application of these regression models will be illustrated by using two numerical data sets.

Keywords: *T-X family, Sarmanov family, nested models, non-nested models*

1 Introduction

Many univariate count data regression models have been defined and studied. Some of these models have been extended to bivariate, and a few of them have been extended to multivariate count data regression models. Examples can be found in the books by Cameron and Trivedi (2013) and Winkelmann (2008) and the references therein.

The major limitation of the standard Poisson regression model is that it can be used to model response count data in which the mean and the variance are about equal, the case of equi-dispersion. Quite often, a count data may be over-dispersed (variance > mean) or under-

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dispersed (variance < mean). Few univariate count data regression models in the literature allow for over- and under-dispersion. Winkelmann (2008, pp. 45-56) discussed three such models. Sellers and Shmueli (2010) gave COM-Poisson regression model. Sun and Ong (2016) discussed the generalized inverse trinomial distribution. These regression models have some drawbacks that include probability mass functions in complicated forms and/or some of the means and variances of the response variables are not available in closed forms.

There are some methods to extend the univariate count data regression models to bivariate or multivariate count data regression models. One method is the trivariate reduction that leads to a bivariate regression model [see Kocherlakota and Kocherlakota, (1992, Chapter 1)]. The drawback of this technique is that the correlation between the two variates is always positive. Modeling multivariate count data using copulas is another common technique. Copula-based models allow for flexible dependence structure and marginal distributions. However, the choice of copulas and the marginal distributions is not an easy task. Nikoloulopoulos and Karlis (2010) considered multivariate Archimedean copulas, partially-symmetric copulas and mixtures of max-id bivariate copulas. Some other types of copula-based multivariate models for count data were reviewed by Nikoloulopoulos (2013) and the references therein. Other techniques to obtain bivariate models and their drawbacks were discussed by Famoye (2010).

In this paper, we define a family of multivariate distributions using the $T-R\{W\}$ framework. The multivariate count data regression models based on the $T-R\{W\}$ family of discrete distributions are defined. Some important characteristics of the multivariate count data regression models include the following: (i) they allow any type of correlation between any two or more variates, (ii) they allow for both under- and over-dispersion for each variate, (iii) they allow correlations and dispersions to be determined independently, and (iv) the cumulative distribution

function (CDF) of each marginal distribution is in closed form. The means and variances of the marginal distributions may not be in closed form, but they are easy to compute.

Another motivation for this work is from the paper by Famoye (2019) on the bivariate exponentiated-exponential geometric regression (BEEGR) model. A simulation study was carried out to compare the BEEGR model with the bivariate generalized Poisson regression (BGPR) model studied by Hofer and Leitner (2012). Data were generated from each of the two regression models, both models were fitted to the data, and the proportion of times one model is significantly better than the other were recorded. P_{ee} denotes the proportion of times BEEGR is better than the BGPR while P_{gp} denotes the proportion of times BGPR is better than the BEEGR. When the data were simulated from BEEGR model, one would expect $P_{ee} > P_{gp}$. One would expect the opposite if the data were simulated from the BGPR model. The results presented by Famoye (2019) showed that P_{ee} exceeded P_{gp} in all cases when the data were generated from the BEEGR model. On the other hand, P_{gp} did not exceed P_{ee} in all cases when the data were simulated from the BGPR model. Based on the conclusion in the paper, the BEEGR model seems to have advantages over the BGPR model.

In [Sect. 2](#), we briefly review the $T-R\{W\}$ family of distributions. In the section, we focus on the $T-R\{W\}$ family of discrete distributions. Some sub-families of the $T-R\{W\}$ discrete distributions are presented with some specific examples. Some examples of generalized geometric distributions are provided. Bivariate and multivariate generalized geometric distributions are defined in [Sect. 3](#). Multivariate generalized geometric regression model is defined in [Sect. 3](#). In [Sect. 4](#), we propose the maximum likelihood method for estimating the model parameters and some tests that include goodness of fit statistics. In [Sect. 5](#), application of

some multivariate generalized geometric regression models is illustrated with two numerical data sets. In [Sect. 6](#), we provide some summary and concluding remarks.

2 Review of $T\text{-}R\{W\}$ family of distributions

There are many methods proposed in the literature for generating statistical distributions. See for example a review by Lee et al. (2013) and the references therein. Alzaatreh et al. (2013) proposed the $T\text{-}X(U)$ family of distributions by connecting the random variable T with support (a, b) and X using the transformation U . The CDF of the $T\text{-}X(U)$ family of distributions is given by

$$G(x) = \int_a^{U(F(x))} r(t)dt = R\{U(F(x))\}, \quad (1)$$

where $U(\cdot): (0, 1) \rightarrow (a, b)$ satisfies the following two conditions: (i) $U(\cdot)$ is monotonically non-decreasing and absolutely continuous, and (ii) $U(0) \rightarrow a$ and $U(1) \rightarrow b$. The CDFs of T and X are $R(t)$ and $F(x)$ respectively. The corresponding probability density function (PDF) for the continuous random variable X in (1) is given by

$$g(x) = f(x)u(F(x))r\{U(F(x))\}, \text{ where } u(x) = \frac{d}{dx}U(x). \quad (2)$$

Giving a function U and a random variable T , the resulting $T\text{-}X(U)$ distribution is a generalized distribution of X .

Aljarrah et al. (2014) considered the function $U(F(x))$ in (1) to be the quantile function $Q_W(F(x))$

$F_T(y) = P(T \leq y)$, $F_R(y) = P(R \leq y)$ and $F_W(y) = P(W \leq y)$ respectively. The corresponding quantile functions are $Q_T(p)$, $Q_R(p)$ and $Q_W(p)$, where $Q_Z(p) = \inf\{z : F_Z(z) \geq p\}$. If the

densities exist, they can be denoted by $f_T(y)$, $f_R(y)$ and $f_W(y)$. The CDF of the random variable Y in the $T-R\{W\}$ family of distributions is defined as

$$F_Y(y) = \int_a^{Q_W(F_R(y))} f_T(t) dt = F_T\{Q_W(F_R(y))\}. \quad (3)$$

It is assumed the supports of the random variables T and W are (a, b) and (c, d) respectively, where $(a, b) \subset (c, d)$ for $-\infty \leq a < b \leq \infty$ and $-\infty \leq c < d \leq \infty$. The PDF corresponding to the CDF in (3) when R is continuous can be written as

$$f_Y(y) = f_R(y) \times \frac{f_T(Q_W(F_R(y)))}{f_W(Q_W(F_R(y)))}. \quad (4)$$

When the support (a, b) of the random variable T is a subset of the support (c, d) of the random variable W , Aljarrah et al. (2014) gave the support for the random variable Y in (3) **when R is continuous**. In this paper, we assume that the support of the random variable T is the same as that of the random variable W . Given the quantile function of random variable W and random variable T , the resulting $T-R\{W\}$ is a generalized distribution of the random variable R . For example, Table 1 shows various $T-R\{W\}$ families based on different choices of Q_W . To simplify the notation, we will use $G(y)$ to replace $F_Y(y)$. Different $T-R\{W\}$ families can be defined using (3), interested readers are referred to Aljarrah et al. (2014), Alzaatreh et al. (2014), Aldeni et al. (2017) and the references therein.

Table 1: Examples of $T-R\{W\}$ distributions based on different choices of random variable W^*

W	$Q_W(p)$	$G(y)$
(a) Uniform	p	$F_T[F_R(y)]$
(b) Exponential	$-\log(1-p)$	$F_T[-\log(1-F_R(y))]$
(c) Log-logistic	$p/(1-p)$	$F_T[F_R(y)/(1-F_R(y))]$
(d) Logistic	$\log(p/(1-p))$	$F_T[\log\{F_R(y)/(1-F_R(y))\}]$

*standard random variable W

If R is a discrete random variable with support $\mathbb{N}_* = \mathbb{N} \cup \{0\}$, then (3) is the CDF of a discrete $T-R\{W\}$ family with the corresponding probability mass function (PMF)

$$g(y) = F_T(Q_W(F_R(y))) - F_T(Q_W(F_R(y-1))), \quad y \in \mathbb{N}_*, \quad (5)$$

where $F(-1) = 0$. An interesting special case of (5) was studied by Alzaatreh et al. (2012). They studied the case T -geometric{standard exponential} which defines a discrete analogue to the non-negative continuous random variable T .

Note that the family of **discrete** distributions in (3) is well defined only if the random variables T and W have the same support. Next, several subfamilies of discrete $T-R\{W\}$ are defined.

2.1 Sub-families of discrete $T-R\{W\}$ distributions

In this sub-section, several sub-families of the discrete $T-R\{W\}$ will be defined. We study the cases when T follows exponential, exponentiated-exponential, Weibull and logistic distributions and W follows standard forms of exponential, log-logistic and logistic distributions.

a. Discrete exponential- $R\{W\}$ distributions:

If a random variable T follows the exponential distribution with parameter λ , then

$f_T(y) = \lambda e^{-\lambda y}$, $y > 0, \lambda > 0$. From (5), the CDF and the PMF of the discrete exponential- $R\{W\}$ are respectively, given by

$$G(y) = 1 - \exp\left\{-\lambda Q_W\left(F_R(\lfloor y \rfloor)\right)\right\}, \quad y \geq 0, \quad (6)$$

$$g(y) = \exp\left\{-\lambda Q_W\left(F_R(\lfloor y-1 \rfloor)\right)\right\} - \exp\left\{-\lambda Q_W\left(F_R(\lfloor y \rfloor)\right)\right\}, \quad y \in \mathbb{N}_*, \quad (7)$$

where $\lfloor y \rfloor = \max\{m \in \mathbb{Z} \mid m \leq y\}$, is the floor function.

Example 1 (Discrete exponential- $R\{\text{standard exponential}\}$ distributions, denoted as E- $R\{\text{Es}\}$):

If Q_W is the quantile function of the exponential distribution given in Table 1, then the CDF of the discrete E- $R\{Es\}$ is given by

$$G(y) = 1 - \left(1 - F_R(\lfloor y \rfloor)\right)^\lambda, \quad y \geq 0. \quad (8)$$

If $\lambda = 1$, $G(y)$ in (8) reduces to $F_R(y)$.

Example 2 (Discrete exponential- $R\{\text{standard log-logistic}\}$ distributions, denoted as E- $R\{LLs\}$):

If Q_W is the quantile function of the standard log-logistic distribution given in Table 1, then the CDF of the discrete E- $R\{LLs\}$ is given by

$$G(y) = 1 - \exp\left(\frac{-\lambda F_R(\lfloor y \rfloor)}{1 - F_R(\lfloor y \rfloor)}\right), \quad y \geq 0. \quad (9)$$

b. Discrete exponentiated exponential- $R\{W\}$ distributions:

If a random variable T follows the exponentiated-exponential distribution defined by Gupta and Kundu (2001) with PDF, $f_T(y) = \alpha\lambda(1 - e^{-\lambda y})^{\alpha-1} e^{-\lambda y}$, $y > 0, \alpha, \lambda > 0$, then the CDF and the PMF of the discrete exponentiated exponential- $R\{W\}$ are respectively, given by

$$G(y) = \left(1 - \exp\left\{-\lambda Q_W\left(F_R(\lfloor y \rfloor)\right)\right\}\right)^\alpha, \quad y \geq 0, \text{ and} \quad (10)$$

$$g(y) = \left(1 - \exp\left\{-\lambda Q_W\left(F_R(\lfloor y - 1 \rfloor)\right)\right\}\right)^\alpha - \left(1 - \exp\left\{-\lambda Q_W\left(F_R(\lfloor y \rfloor)\right)\right\}\right)^\alpha, \quad y \in \mathbb{N}_*. \quad (11)$$

Example 3 (Discrete exponentiated exponential- $R\{\text{standard exponential}\}$ distributions, denoted as EE- $R\{Es\}$):

If Q_W is the quantile function of the standard exponential distribution given in Table 1, then the CDF of the discrete EE- $R\{Es\}$ is given by

$$G(y) = \left\{1 - \left(1 - F_R(\lfloor y \rfloor)\right)^\lambda\right\}^\alpha, \quad y \geq 0. \quad (12)$$

If $\alpha = 1$, $G(y)$ in (12) reduces to the CDF of the discrete exponential- R {standard exponential} in (8).

Example 4 (Discrete exponentiated exponential- R {standard log-logistic} distributions, denoted as EE- R {LLs}):

If Q_W is the quantile function of the standard log-logistic distribution given in Table 1, then the CDF of the discrete EE- R {LLs} is given by

$$G(y) = \left\{ 1 - \exp \left(\frac{-\lambda F_R(\lfloor y \rfloor)}{1 - F_R(\lfloor y \rfloor)} \right) \right\}^\alpha, \quad y \geq 0. \quad (13)$$

If $\alpha = 1$, $G(y)$ in (13) reduces to the CDF of the discrete exponential- R {log-logistic} in (9).

c. Discrete Weibull- R { W } distributions:

If a random variable T follows the Weibull distribution with parameters c and λ , then

$f_T(y) = c\lambda^c y^{c-1} e^{-(\lambda y)^c}$, $y \geq 0$. The CDF and the PMF of the discrete Weibull- R { W } are respectively, given by

$$G(y) = 1 - \exp \left\{ - \left(\lambda Q_W \left(F_R(\lfloor y \rfloor) \right) \right)^c \right\}, \quad y \geq 0, \quad (14)$$

$$g(y) = \exp \left\{ - \left(\lambda Q_W \left(F_R(\lfloor y - 1 \rfloor) \right) \right)^c \right\} - \exp \left\{ - \left(\lambda Q_W \left(F_R(\lfloor y \rfloor) \right) \right)^c \right\}, \quad y \in \mathbb{N}_*. \quad (15)$$

Example 5 (Discrete Weibull- R {standard exponential} distributions, denoted as W- R {Es}):

If Q_W is the quantile function of the standard exponential distribution given in Table 1, then the CDF of the discrete W- R {Es} is given by

$$G(y) = 1 - \exp \left\{ - \left(-\lambda \log(1 - F_R(\lfloor y \rfloor)) \right)^c \right\}, \quad y \geq 0. \quad (16)$$

If $c = 1$, $G(y)$ in (16) reduces to the CDF of E-R{Es} in (8).

Example 6 (Discrete Weibull-R{standard log-logistic} distributions, denoted as W-R{LLs}):

If Q_W is the quantile function of the standard log-logistic distribution given in Table 1, then the CDF of the discrete W-R{LLs} is given by

$$G(y) = 1 - \exp \left\{ - \left(\frac{\lambda F_R(\lfloor y \rfloor)}{1 - F_R(\lfloor y \rfloor)} \right)^c \right\}, \quad y \geq 0. \quad (17)$$

If $c = 1$, $G(y)$ in (17) reduces to the CDF of the discrete E-R{LLs} in (9).

d. Discrete logistic-R{ W } distributions:

If a random variable T follows the logistic distribution with parameter λ , then

$f_T(y) = \lambda e^{-\lambda y} (1 + e^{-\lambda y})^{-2}$, $-\infty < y < \infty$. The CDF and the PMF of the discrete L-R{ W } are respectively, given by

$$G(y) = \frac{1}{1 + \exp \left\{ -\lambda Q_W \left(F_R(\lfloor y \rfloor) \right) \right\}}, \quad y \geq 0 \quad \text{and} \quad (18)$$

$$g(y) = \frac{1}{1 + \exp \left\{ -\lambda Q_W \left(F_R(\lfloor y \rfloor) \right) \right\}} - \frac{1}{1 + \exp \left\{ -\lambda Q_W \left(F_R(\lfloor y - 1 \rfloor) \right) \right\}}, \quad y \in \mathbb{N}_*. \quad (19)$$

Example 7 (Discrete logistic-R{standard logistic} distributions, denoted as L-R{Ls}):

If Q_W is the quantile function of the logistic distribution given in Table 1, then the CDF of the discrete L-R{Ls} family is given by

$$G(y) = \frac{F_R^\lambda(\lfloor y \rfloor)}{F_R^\lambda(\lfloor y \rfloor) + (1 - F_R(\lfloor y \rfloor))^\lambda}, \quad y \geq 0. \quad (20)$$

If $\lambda = 1$, $G(y)$ in (20) reduces to the baseline CDF $F_R(y)$.

2.2 The family of generalized geometric distributions

In this sub-section, we propose a family of generalized geometric distributions by taking the random variable R in the discrete $T-R\{W\}$ framework to follow the geometric distribution with the CDF $F_R(y) = 1 - \varphi^{y+1}$, $0 < \varphi < 1$, $y = 0, 1, 2, \dots$. The random variable Y denotes the number of failures to obtain the first success and φ is the probability of a failure. Table 2 provides the CDF of some examples of T -geometric $\{W\}$ families. The corresponding PMFs can be found using equations (7), (11), (15) or (19).

Table 2: T -geometric $\{W\}$ families based on different choices for the random variables W and T

T -geometric $\{W\}$	CDF $G(y)$
(i) E-geometric $\{Es\}$	$G(y) = 1 - \theta^{y+1}$; $\theta \in (0, 1)$
(ii) E-geometric $\{LLs\}$	$G(y) = 1 - \exp\left\{-\lambda\left(\theta^{-(y+1)} - 1\right)\right\}$; $\lambda > 0, \theta \in (0, 1)$
(iii) EE-geometric $\{Es\}$	$G(y) = (1 - \theta^{y+1})^\alpha$; $\alpha > 0, \theta \in (0, 1)$
(iv) EE-geometric $\{LLs\}$	$G(y) = \left(1 - \exp\left\{-\lambda\left(\theta^{-(y+1)} - 1\right)\right\}\right)^\alpha$; $\lambda, \alpha > 0, \theta \in (0, 1)$
(v) W-geometric $\{Es\}$	$G(y) = 1 - \theta^{(y+1)^c}$; $c > 0, \theta \in (0, 1)$
(vi) W-geometric $\{LLs\}$	$G(y) = 1 - \exp\left\{-\left(\lambda(\theta^{-(y+1)} - 1)\right)^c\right\}$; $\lambda, c > 0, \theta \in (0, 1)$
(vii) L-geometric $\{Ls\}$	$G(y) = (1 - \theta^{y+1})^\lambda / [(1 - \theta^{y+1})^\lambda + \theta^{\lambda(y+1)}]$; $\lambda > 0, \theta \in (0, 1)$

Remarks:

- (a) Note that the random variable Y takes values $y = 0, 1, 2, \dots$ for all distributions in Table 2.
- (b) Exponential-geometric{standard exponential} in Table 2(i) is just the geometric distribution and exponentiated exponential-geometric{standard exponential} in Table 3(iii) is the exponentiated exponential-geometric distribution defined by Alzaatreh et al. (2012). By

$$G(y) = 1 - \left(1 - F_R(\lfloor y \rfloor)\right)^\lambda = 1 - \varphi^{\lambda(y+1)} = 1 - \theta^{y+1}, \text{ where } \theta = \varphi^\lambda. \text{ See}$$

Proposition 1 in Lee et al. (2021), where it was shown that the CDF of T -geometric family is

characterized by $\theta^{\xi(y)}$ for $\theta \in (0, 1)$, where ξ is a function of y . For the Weibull-

geometric {standard exponential}, we use (16) to obtain $G(y) = 1 - \exp\left\{-\left(-\lambda \log \varphi^{y+1}\right)^c\right\} =$

$1 - \exp\left\{-(-\lambda \log \varphi)^c \times (y+1)^c\right\} = 1 - [e^{-(-\lambda \log \varphi)^c}]^{(y+1)^c} = 1 - \theta^{(y+1)^c}$, where $\theta = e^{-(-\lambda \log \varphi)^c}$. Some

of the probability mass functions in Table 2 are derived in Chapter Appendix.

- (c) All the families of discrete distributions generated in this section can also be generated by using any other discrete random variable R with CDF $F_R(y)$. For example, the random variable R can be replaced by the Poisson, binomial, and negative binomial distributions. The only difference is that the geometric distribution has a closed form CDF. An option is to use approximate functions (e.g. incomplete gamma function for Poisson) to calculate the CDF.
- (d) By using the exponentiated Weibull distribution in place of the Weibull distribution, one will obtain the exponentiated Weibull-geometric distribution studied by Famoye (2019).
- (e) These new discrete distributions are characterized by at least one or more new parameters. For example in Table 2, all the distributions in 2(ii) through 2(vii) are generalization of the geometric distribution characterized by one parameter. Among the new parameters is a shape parameter or a scale parameter. Only the distribution in 2(ii) has new scale parameter. All others from 2(iii) through 2(vii) have either one new shape parameter or have one new shape parameter and one new scale parameter. Of course, one can have more parameters if the standard quantile functions are not used. These generalized new distributions are derived to have one or two more parameters than the commonly used classic distributions. Thus, the additional parameters are practically useful for capturing the shape and/or scale of the

empirical data which could not be properly fitted by the classic distributions. This is one way to reduce the number of parameters and provide enough flexibility for fitting empirical data in the derived new distributions. Furthermore, one can apply test statistics given in Sect. 4 to test if the new parameters are statistically significant to select the most appropriate distribution for fitting a given data.

Bivariate and multivariate extensions of T -geometric $\{W\}$ family is considered in the next section.

3 Bivariate and multivariate T -geometric $\{W\}$ families

In this section, we use the technique discussed in Sarmanov (1966) to propose bivariate and multivariate extensions of univariate T -geometric $\{W\}$ family. Lee (1996) extended bivariate Sarmanov distribution to multivariate case with several parameters measuring covariances of order 2, 3, ..., for m -variate distribution.

3.1 Sarmanov family of bivariate and multivariate distributions

In this sub-section, we define the bivariate and multivariate Sarmanov family of distributions.

Definition 1 (Bivariate Sarmanov family of distributions):

Let Y_1 and Y_2 be two random variables with their probability density (or mass) functions $f_1(y_1)$ and $f_2(y_2)$. Let $\phi_t(u)$, $t = 1, 2$ be bounded nonconstant functions such that $E[\phi_t(Y_t)] = 0$. Then the bivariate Sarmanov family is defined as

$$h(y_1, y_2) = f_1(y_1)f_2(y_2)\{1 + \omega\phi_1(y_1)\phi_2(y_2)\}, \quad (y_1, y_2) \in \mathbb{R} \times \mathbb{R}, \quad (21)$$

where $\omega \in \mathbb{R}$ satisfies the condition $1 + \omega\phi_1(y_1)\phi_2(y_2) \geq 0$ for all y_1 and y_2 .

Theorem 1 Assume (Y_1, Y_2) follows the Sarmanov family in (21) with μ_t and σ_t^2 the mean and variance for Y_t , $t = 1, 2$. Then

- (i) The marginal probability mass functions in (21) are $f_1(y_1)$ and $f_2(y_2)$.
- (ii) If $\omega = 0$, then Y_1 and Y_2 are independent.
- (iii) $E(Y_1 Y_2) = \mu_1 \mu_2 + \omega v_1 v_2$, where $v_t = E[Y_t \phi_t(Y_t)]$, $t = 1, 2$.
- (iv) $E(Y_2 | Y_1 = y_1) = \mu_2 + \omega v_2 \phi_1(y_1)$.
- (v) $\rho = \text{corr}(Y_1, Y_2) = \frac{\omega v_1 v_2}{\sigma_1 \sigma_2}$.

Proof: Straightforward from (21). For more details refer to Lee (1996).

Remark 2: Let Y_1 and Y_2 be two random variables defined on $[0, \infty)$ with moment generating functions $M_t(u) = E(e^{uY_t})$, $t = 1, 2$. Define $\phi_t(y_t) = e^{-y_t} - M_t(-1)$. Then $h(y_1, y_2)$ in (21) is a valid joint probability density (or mass) function.

Definition 2 (Multivariate Sarmanov family of distributions):

Let $Y_1, Y_2, \dots, Y_d$ be random variables with their probability density or mass functions $f_1, f_2, \dots, f_d$.

Let $\phi_t(y)$, $t = 1, 2, \dots, d$ be bounded nonconstant functions such that $E[\phi_t(Y_t)] = 0$. Then the multivariate Sarmanov family is defined as

$$h(y_1, y_2, \dots, y_d) = \left\{ \prod_{t=1}^d f_t(y_t) \right\} \left(1 + R_{\phi_1, \phi_2, \dots, \phi_d, \Omega_d}(y_1, y_2, \dots, y_d) \right), \quad (y_1, y_2, \dots, y_d) \in \mathbb{R}^d, \quad (22)$$

where $R_{\phi_1, \phi_2, \dots, \phi_d, \Omega_d}(y_1, y_2, \dots, y_d) = \sum_{j_1 < j_2}^d \omega_{j_1, j_2} \phi_{j_1} \phi_{j_2} + \sum_{j_1 < j_2 < j_3}^d \omega_{j_1, j_2, j_3} \phi_{j_1} \phi_{j_2} \phi_{j_3} + \dots + \omega_{1, 2, \dots, d} \prod_{t=1}^d \phi_t$ and

$\Omega_d = \{\omega_{j_1, j_2}, \omega_{j_1, j_2, j_3}, \dots, \omega_{1, 2, \dots, d}\}$ is chosen so that $R_{\phi_1, \phi_2, \dots, \phi_d, \Omega_d}(y_1, y_2, \dots, y_d) \geq -1$.

Theorem 1 can be extended to the multivariate case. For more details, see Lee (1996). Next, bivariate, and multivariate extensions of the T -geometric $\{W\}$ family are proposed.

3.2 Bivariate and multivariate T -geometric $\{W\}$ families

Let Y_1 and Y_2 follow the discrete T - $R\{W\}$ family in (5) with PDF $f_1(y_1)$ and $f_2(y_2)$ respectively.

Define $c_t = M_t(-1)$ where $M_t(u)$ is the moment generating function of Y_t , $t = 1, 2$. Then by

Remark 2, the bivariate discrete T - $R\{W\}$ family can be written as

$$h(y_1, y_2) = f_1(y_1)f_2(y_2)\{1 + \omega(e^{-y_1} - c_1)(e^{-y_2} - c_2)\}, \quad (y_1, y_2) \in \mathbb{N}_* \times \mathbb{N}_*, \quad (23)$$

Table 3 presents several examples of bivariate discrete T - $R\{W\}$ family. In Table 3

$$\delta(y_1, y_2) = \omega(e^{-y_1} - c_1)(e^{-y_2} - c_2).$$

Table 3: Bivariate T -geometric $\{W\}$ families based on different choices for variables W and T

Bivariate T -geometric $\{W\}$	Joint probability mass function $h(y_1, y_2)$
(i) BE-geometric $\{Es\}$	$\prod_{t=1}^2 [\theta_t^{y_t} (1 - \theta_t)] \times \{1 + \delta(y_1, y_2)\}$
(ii) BE-geometric $\{LLs\}$	$\prod_{t=1}^2 \left[e^{-\lambda_t(\theta_t^{-y_t} - 1)} - e^{-\lambda_t(\theta_t^{-(y_t+1)} - 1)} \right] \times \{1 + \delta(y_1, y_2)\}$
(iii) BEE-geometric $\{Es\}$	$\prod_{t=1}^2 \left[(1 - \theta_t^{y_t+1})^{\alpha_t} - (1 - \theta_t^{y_t})^{\alpha_t} \right] \times \{1 + \delta(y_1, y_2)\}.$
(iv) BEE-geometric $\{LLs\}$	$\prod_{t=1}^2 \left[\left(1 - e^{-\lambda_t(\theta_t^{-(y_t+1)} - 1)} \right)^{\alpha_t} - \left(1 - e^{-\lambda_t(\theta_t^{-y_t} - 1)} \right)^{\alpha_t} \right] \times \{1 + \delta(y_1, y_2)\}$
(v) BW-geometric $\{Es\}$	$\prod_{t=1}^2 [\theta_t^{(y_t+1)^{c_t}} - \theta_t^{y_t^{c_t}}] \times \{1 + \delta(y_1, y_2)\}$
(vi) BW-geometric $\{LLs\}$	$\prod_{t=1}^2 \left[e^{-\left(\lambda_t(\theta_t^{-(y_t+1)} - 1)\right)^{c_t}} - e^{-\left(\lambda_t(\theta_t^{-y_t} - 1)\right)^{c_t}} \right] \times \{1 + \delta(y_1, y_2)\}$
(vii) BL-geometric $\{Ls\}$	$\prod_{t=1}^2 \left[\frac{(1 - \theta_t^{y_t+1})^{\lambda_t}}{(1 - \theta_t^{y_t+1})^{\lambda_t} + \theta_t^{\lambda_t(y_t+1)}} - \frac{(1 - \theta_t^{y_t})^{\lambda_t}}{(1 - \theta_t^{y_t})^{\lambda_t} + \theta_t^{\lambda_t y_t}} \right] \times \{1 + \delta(y_1, y_2)\}$

A multivariate extension of the discrete $T\text{-}R\{W\}$ family can be defined using (21). For

simplicity, let $R_{\phi_1, \phi_2, \dots, \phi_d, \Omega_d}(y_1, y_2, \dots, y_d) = \sum_{j_1 < j_2}^d \omega_{j_1, j_2} \phi_{j_1} \phi_{j_2}$ where $\phi_t(y_t) = e^{-y_t} - c_t$ and

$c_t = M_t(-1)$, $t = 1, 2, \dots, d$. Then, from (22) and (5), the multivariate discrete $T\text{-}R\{W\}$ family is given by

$$h(y_1, y_2, \dots, y_d) = \prod_{t=1}^d [f_t(y_t)] \times \left\{ 1 + \sum_{t < v} \omega_{tv} (e^{-y_t} - c_t)(e^{-y_v} - c_v) \right\}, \quad (y_1, y_2, \dots, y_d) \in \mathbb{N}_*^d. \quad (24)$$

Table 4 presents several examples of multivariate $T\text{-geometric}\{W\}$ family. In Table 4

$$\delta(x_1, x_2, \dots, x_d) = \sum_{t < v}^d \omega_{tv} (e^{-x_t} - c_t)(e^{-x_v} - c_v). \text{ Note that a multivariate version with more than}$$

pairwise associations between the response variables can be defined.

Table 4: Multivariate $T\text{-geometric}\{W\}$ families based on different choices for variables W and T

Multivariate $T\text{-geometric}\{W\}$	Joint probability mass function $h(y_1, y_2, \dots, y_d)$
(i) ME-geometric{Es}	$\prod_{t=1}^d [\theta_t^{y_t} (1 - \theta_t)] \times \{1 + \delta(y_1, y_2, \dots, y_d)\}$
(ii) ME-geometric{LLs}	$\prod_{t=1}^d \left[e^{-\lambda_t (\theta_t^{y_t} - 1)} - e^{-\lambda_t (\theta_t^{-(y_t+1)} - 1)} \right] \times \{1 + \delta(y_1, y_2, \dots, y_d)\}$
(iii) MEE-geometric{Es}	$\prod_{t=1}^d \left[(1 - \theta_t^{y_t+1})^{\alpha_t} - (1 - \theta_t^{y_t})^{\alpha_t} \right] \times \{1 + \delta(y_1, y_2, \dots, y_d)\}$
(iv) MEE-geometric{LLs}	$\prod_{t=1}^d \left[\left(1 - e^{-\lambda_t (\theta_t^{-(y_t+1)} - 1)} \right)^{\alpha_t} - \left(1 - e^{-\lambda_t (\theta_t^{y_t} - 1)} \right)^{\alpha_t} \right] \times \{1 + \delta(y_1, y_2, \dots, y_d)\}$
(v) MW-geometric{Es}	$\prod_{t=1}^d [\theta_t^{(y_t+1)^{c_t}} - \theta_t^{y_t^{c_t}}] \times \{1 + \delta(y_1, y_2, \dots, y_d)\}$
(vi) MW-geometric{LLs}	$\prod_{t=1}^d \left[e^{-\left(\lambda_t (\theta_t^{y_t+1} - 1) \right)^{c_t}} - e^{-\left(\lambda_t (\theta_t^{-(y_t+1)} - 1) \right)^{c_t}} \right] \times \{1 + \delta(y_1, y_2, \dots, y_d)\}$
(vii) ML-geometric{Ls}	$\prod_{t=1}^d \left[\frac{(1 - \theta_t^{y_t+1})^{\lambda_t}}{(1 - \theta_t^{y_t+1})^{\lambda_t} + \theta_t^{\lambda_t (y_t+1)}} - \frac{(1 - \theta_t^{y_t})^{\lambda_t}}{(1 - \theta_t^{y_t})^{\lambda_t} + \theta_t^{\lambda_t y_t}} \right] \{1 + \delta(y_1, \dots, y_d)\}$

3.3 Multivariate T -geometric $\{W\}$ regression model

Let the count response variable Y_{it} , $i = 1, 2, \dots, n$; $t = 1, 2, \dots, d$ where n is the sample size and let

$\mathbf{x}_i = (x_{i0} = 1, x_{i1}, x_{i2}, \dots, x_{i(k-1)})^T$ be a vector of $k-1$ covariates. We assume that the same

covariates affect each count response $\mathbf{Y}_i$, thus, $\mathbf{x}_{i1} = \mathbf{x}_{i2} = \dots = \mathbf{x}_{id} = \mathbf{x}_i$. Assume that the

conditional distribution of $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, \dots, Y_{id})^T$ for any given $\mathbf{x}_i = (x_{i0}, x_{i1}, x_{i2}, \dots, x_{i(k-1)})^T$,

follows the multivariate T -geometric $\{W\}$ family with

$E(Y_{it} | x_{ij}) = \theta_t(x_{ij}) = \gamma(x_{ij}, \beta_{ij})$, $t = 1, 2, \dots, d$, and $j = 0, 1, \dots, (k-1)$, where $0 < \gamma(\mathbf{x}_i, \boldsymbol{\beta}_t) < 1$ is a

known function of x_{ij} , and a vector $\boldsymbol{\beta}_t = (\beta_{t0}, \beta_{t1}, \dots, \beta_{t(k-1)})$ of regression parameters. Now, let

the function $\gamma(\mathbf{x}_i, \boldsymbol{\beta}_t)$ be the logit function. Therefore,

$$\theta_t(\mathbf{x}_i) = \theta_t = \gamma(\mathbf{x}_i, \boldsymbol{\beta}_t) = \frac{\exp(\mathbf{x}_i^T \boldsymbol{\beta}_t)}{1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta}_t)}. \quad (25)$$

This implies that the bivariate and multivariate T -geometric $\{W\}$ regression models can be

written respectively as

$$h(y_{i1}, y_{i2} | \mathbf{x}_i) = f_1(y_{i1}; \theta_1) f_2(y_{i2}; \theta_2) \{1 + \omega(e^{-y_{i1}} - c_1)(e^{-y_{i2}} - c_2)\}, \quad (26)$$

$$h(y_{i1}, y_{i2}, \dots, y_{id}) = \prod_{t=1}^d [f_t(y_{it}; \theta_t)] \times \left\{ 1 + \sum_{t < v} \omega_{tv} (e^{-y_{it}} - c_t)(e^{-y_{iv}} - c_v) \right\}, \quad (27)$$

where θ_t is given in (25) and $f_t(y_{it}; \theta_t)$ follows T -geometric $\{W\}$ family for $t = 1, 2, \dots, d$. In the

next sections, we will discuss the statistical inference for the bivariate T -geometric $\{W\}$

regression model and application to real life data sets. All results in these sections can be

extended to the multivariate case in (27).

4 Inference on bivariate and multivariate T -geometric $\{W\}$ regression models

The maximum likelihood estimation method can be used to estimate the unknown parameters in the bivariate T -geometric $\{W\}$ regression model in (26). If a random sample of size n is taken from the bivariate T -geometric $\{W\}$ regression model in (26), then the log-likelihood function is given by

$$\begin{aligned}\log L(\boldsymbol{\beta}, \boldsymbol{\alpha}) &= \sum_{i=1}^n \log h(y_{i1}, y_{i2} | \mathbf{x}_i) \\ &= \sum_{i=1}^n \sum_{t=1}^2 \log f_t(y_{it}; \theta_t) + \sum_{i=1}^n \log \{1 + \omega(e^{-y_{i1}} - c_1)(e^{-y_{i2}} - c_2)\},\end{aligned}\quad (28)$$

where $\boldsymbol{\beta}$ is the vector of all unknown regression parameters and $\boldsymbol{\alpha}$ is the vector of parameters from the marginal PMFs in $h(y_{i1}, y_{i2} | \mathbf{x}_i)$. In next sub-sections, we discuss some statistical tests and goodness of fit measures.

By using the discrete exponentiated-exponential geometric distribution in (11) with φ^λ set to parameter θ , we obtain the multivariate exponentiated-exponential geometric regression (MEEGR) model as

$$P(y_1, y_2, \dots, y_d) = \prod_{t=1}^d \left[(1 - \theta_t^{y_t+1})^{b_t} - (1 - \theta_t^{y_t})^{b_t} \right] \left[1 + \sum_{t < v} \omega_v (e^{-y_t} - c_t)(e^{-y_v} - c_v) \right] \quad (29)$$

where $\theta_t = \theta_t(x_i) = \exp(\beta_{t0} + \beta_{t1}x_{i1}) / [1 + \exp(\beta_{t0} + \beta_{t1}x_{i1})] = 1 / \{1 + \exp[-(\beta_{t0} + \beta_{t1}x_{i1})]\}$. The marginal PMF for each y_t is $P(y_t) = (1 - \theta_t^{y_t+1})^{b_t} - (1 - \theta_t^{y_t})^{b_t}$. Suppose $d = 3$, the conditional PMF of y_2 given y_1 is

$$P(y_2 | y_1) = [(1 - \theta_2^{y_2+1})^{b_2} - (1 - \theta_2^{y_2})^{b_2}] [1 + \omega_{12}(e^{-y_1} - c_1)(e^{-y_2} - c_2)] / y_2! \quad (30)$$

The conditional PMF of y_3 given y_1 and y_2 is given by

$$P(y_3 | y_1, y_2) = [(1 - \theta_3^{y_3+1})^{b_3} - (1 - \theta_3^{y_3})^{b_3}] [1 + A / \{1 + \omega_{12}(e^{-y_1} - c_1)(e^{-y_2} - c_2)\}], \quad (31)$$

where $A = \omega_{13}(e^{-y_1} - c_1)(e^{-y_3} - c_3) + \omega_{23}(e^{-y_2} - c_2)(e^{-y_3} - c_3)$. These results may be used when simulating from the multivariate MEEG distribution. Other multivariate generalized geometric distributions can be derived by using the different methods suggested in Table 4. When the dispersion parameters $b_t, t = 1, 2, \dots, d$ are all set to 1, the MEEGR in (29) reduces to the multivariate geometric regression (MGR) model. The initial estimates for the MEEGR model can be taken as the final estimates from the univariate EEGR models for the responses. The initial estimates of the correlations can be the sample correlations.

4.1 Test for independence

A pair of random variables Y_t and Y_v are independent when the parameter ω_{tv} (or ρ_{tv}) is equal to zero. For a multivariate regression model $h(y_{i1}, y_{i2}, \dots, y_{id})$, we test the null hypothesis

$$H_0 : \omega_{tv} = 0 \text{ against } H_a : \omega_{tv} \neq 0, \text{ where } t < v, t = 1, 2, 3, \dots, d, \text{ and } v = 2, 3, \dots, d.$$

Suppose L_{ind} is the likelihood function under H_0 and L_a is the likelihood function under H_a .

The test statistic is given by $\chi_{ind}^2 = -2 \log(L_{ind} / L_a)$, which has a chi-square distribution with $d(d-1)/2$ degrees of freedom.

4.2 Test for dispersion

The MEEGR model in (29) reduces to the MGR model when the dispersion parameters $b_t = 1$, for $t = 1, 2, \dots, d$. To carry out this test, we test if the $b_t = 1$. This gives a situation where all the response variables are equi-dispersed. Thus, we test the hypotheses

$$H_0 : b_t = 1 \text{ for } t = 1, 2, \dots, d \text{ against } H_a : H_0 \text{ is not true.}$$

If L_{dis} is the likelihood function when H_0 is true and L_a is the likelihood function when H_0 is false, the test statistic is given by $\chi_{dis}^2 = -2 \log(L_{dis} / L_a)$, which has an approximate chi-square distribution with d degrees of freedom.

4.3 Test to compare nested and non-nested models

In this sub-section, we use the notation $\underline{y}_i = y_{i1}, y_{i2}, \dots, y_{id}$ and compare the model $f(\underline{y}_i)$ (e.g. MEEGR) and model $g(\underline{y}_i)$ (e.g. MGPR defined by Famoye (2015)). Given the two regression models, we consider the following hypotheses:

$$\begin{aligned} H_0 : & \text{Model } f(\underline{y}_i) \text{ and model } g(\underline{y}_i) \text{ are equivalent} \\ \text{against} \end{aligned} \tag{32}$$

$$H_f : f(\underline{y}_i) \text{ is better than } g(\underline{y}_i) \text{ or } H_g : g(\underline{y}_i) \text{ is better than } f(\underline{y}_i).$$

The likelihood ratio statistic for testing model $f(\underline{y}_i)$ against model $g(\underline{y}_i)$ is defined as

$$L_* = \sum_{i=1}^n \log \left(\frac{f(\underline{y}_i)}{g(\underline{y}_i)} \right). \tag{33}$$

If the two models $f(\underline{y}_i)$ and $g(\underline{y}_i)$ are nested, the statistic in (33) has a chi-square distribution.

If the two models are not nested, the statistic in (33) is not chi-square distributed.

Vuong (1989) used the Kullback-Liebler information criterion to discriminate between two non-nested models. To test the null hypothesis H_0 in (32), Vuong proposed the test statistic

$$Z_* = \frac{L_*}{\hat{w}\sqrt{n}}, \text{ where } w^2 = \frac{1}{n} \sum_{i=1}^n \left[\log \left(\frac{f(\underline{y}_i)}{g(\underline{y}_i)} \right) \right]^2 - \left[\frac{1}{n} \sum_{i=1}^n \log \left(\frac{f(\underline{y}_i)}{g(\underline{y}_i)} \right) \right]^2 \tag{34}$$

is an estimate of the variance of $L_* / \sqrt{n}$. For a non-nested model, Vuong showed that Z_* is approximately standard normal distributed under H_0 in (32). At significant level α , H_0 is rejected in favor of H_g if $Z_* < -z_{\alpha/2}$, and H_0 is rejected in favor of H_f if $Z_* > z_{\alpha/2}$. If $|Z_*| \leq z_{\alpha/2}$, we fail to reject H_0 . Thus, both models are equivalent.

4.4 Goodness-of-fit statistics

The log-likelihood defined in (26) for the bivariate regression model can be used as a goodness of fit statistics for the model. Other alternative measures of goodness of fit are the AIC and the BIC. These statistics are based on the log-likelihood statistic and they are defined as follows: $AIC = -2LL + 2p$, $BIC = -2LL + p \log(n)$, where n is the sample size, p is the number of estimated parameters, and LL is the log-likelihood statistic. Both the AIC and BIC are better than the log-likelihood because they consider the number of parameters in the regression model to control over-parametrization. These measures are usually provided by the SAS NLMIXED procedure. The smaller the measures, the better the model.

5 Application

In this section, we illustrate the applications of MEEGR model to two real life data sets. The results from MEEGR are compared with the results from the MGPR model. The first data set is on the number of sex partners and the second data set is on profiling inmates. For each application, we assume the marginal mean of each Y_{it} in the MGPR model has a log-linear relationship with the covariates x_{ij} ($j = 1, 2, \dots, k$) through

$$\log[E(Y_{it})] = \beta_{t0} + \beta_{t1}x_{it1} + \dots + \beta_{tk}x_{itk}, \text{ where } i = 1, 2, \dots, n \text{ and we have } k \text{ covariates.}$$

The probability $\theta_t = \theta_t(x_{itj})$ in the MEEGR is defined by the logit function

$$\theta_t(x_{it}) = 1 / \{1 + \exp[-(\beta_{t0} + \beta_{t1}x_{it1} + \dots + \beta_{tk}x_{itk})]\}, \text{ where } i = 1, 2, \dots, n \text{ for } k \text{ covariates.}$$

5.1 Sex Partners Data

The number of sex partners data was used by Famoye (2015) to illustrate the multivariate generalized Poisson regression model. See the reference for more information about the data.

Famoye (2015) considered the following three response variables: y_1 = number of male sex partners in the past year, y_2 = number of male oral sex partners in the past year, and y_3 = number of sex partners in the past year who are five years older than the respondent. The descriptive statistics and the pairwise correlations between the response variables are presented in Table 5. Both variables y_1 and y_2 are under-dispersed since the sample variances are smaller than their corresponding sample means. The predictor variables used by Famoye (2015) were race (1 = white, 0 = others), educational level (range from 1 = below 9th grade to 5 = college graduate), marital status (1 = married, 0 = others), and age.

Table 5: Pairwise correlations, mean and variance for response variables in sex partners' data

	Correlation		Mean	Variance
	y_2	y_3		
y_1	0.1345*	0.3581*	1.3133	1.0847
y_2		0.1078*	0.3719	0.3495
y_3			0.3672	0.4108

* significant at 5% level ($n = 1280$)

The data is fitted to the MEEGR model and the model fit is compared with that of MGPR model. The results are presented in Table 6. The loglikelihood value from the MEEGR and

to be 23.24 (p -value < 0.0001), which is highly significant. This indicates that MEEGR model provides a better fit than the MGPR model. The effects of the explanatory variables on the response variables for MGPR model is the same for the MEEGR model. Thus, the response variables y_1 and y_3 are negatively affected by marital status and age. The response variable y_2 is negatively affected by race, education, and age. The older respondents tend to have smaller number of male sex partners, smaller number of oral sex partners, and smaller number of sex partners five years older during the past year. On the relationship between the response variables, ordinary correlation shows that they are all positively correlated and significant. The MGPR and the MEEGR models show that variables y_1 and y_3 have significant positive correlation.

Table 6: Parameter estimates (standard errors in parentheses) for sex partners' data

	y_1		y_2	
Variable	MEEGR model	MGPR model	MEEGR model	MGPR model
Constant (x_0)	-1.515 (0.174)*	0.735 (0.108)*	-1.271 (0.263)*	0.232 (0.206)
Race (x_1)	0.006 (0.088)	0.005 (0.049)	-0.478 (0.105)*	-0.433 (0.104)*
Education (x_2)	0.031 (0.033)	0.022 (0.021)	-0.141 (0.040)*	-0.121 (0.039)*
Marital (x_3)	-0.577 (0.097)*	-0.383 (0.050)*	-0.041 (0.103)	-0.070 (0.103)
Age (x_4)	-0.013 (0.003)*	-0.009 (0.002)*	-0.020 (0.005)*	-0.018 (0.005)*
$\hat{b}_t$	32.411 (3.962)*	-0.028 (0.007)*	6.454 (1.143)*	-0.044 (0.027)

	y_3	
Variable	MEEGR model	MGPR model
Constant (x_0)	-0.931 (0.267)*	-0.160 (0.267)
Race (x_1)	0.018 (0.111)	0.021 (0.120)
Education (x_2)	-0.051 (0.045)	-0.047 (0.051)
Marital (x_3)	-0.646 (0.115)*	-0.774 (0.132)*
Age (x_4)	-0.014 (0.005)*	-0.017 (0.006)*
$\hat{b}_t$	2.831 (0.402)*	0.142 (0.062)*
$\hat{\rho}_{tv}$ for MEEGR	$\hat{\rho}_{12} = 0.0272 (0.0360)$	

	$\hat{\rho}_{13} = 0.2552 (0.0378)^*$ $\hat{\rho}_{23} = 0.0241 (0.0282)$
$\hat{\rho}_{iv}$ for MGPR	$\hat{\rho}_{12} = 0.1009 (0.0735)$ $\hat{\rho}_{13} = 0.5574 (0.0843)^*$ $\hat{\rho}_{23} = 0.0352 (0.0330)$
Log-likelihood	For MEEGR: -2926.21 For MGPR: -3520.63

* significant at 5% level

5.2 Inmates Profiling Data

By 1996, the Los Angeles (LA) county jails faced a serious overcrowding problem. Two suggested solutions to the problem were to build more jail capacity or to divert a greater number of incoming inmates to community-based, intermediate sanctions. A research team was formed to review a profile of inmates in the jail system and to determine how many of them were suitable candidates for intermediate sanctions. A random sample of 1000 un-convicted inmates were selected. Demographic variables include age, race, current offense severity, and gender. The sample size for the collected data set is 931. Among the other variables collected are prior arrests (y_1), number of subsequent arrests (y_2), number of LA custodies (y_3) and number of CDC custodies (y_4). See the reference Petersilia et al. (1996) for information about the data.

In the collected data, the race variable consists of Black, Hispanic, White and others with a frequency of 30. We define two dummy variables x_4 and x_5 to represent the groups Black and Hispanic respectively. Thus, the White group is the reference category. The other explanatory variables are x_1 , x_2 , and x_3 representing age, rank (current offense severity), and gender (where male is represented by 1) respectively. After eliminating any missing case for the four response variables and the five explanatory variables, we are left with a total of $n = 781$. In this sample, the percentages of male, black, Hispanic, and white are respectively 94%, 40%, 45% and 15%

respectively. Table 7 contains the descriptive statistics for the response variables and their bivariate correlations.

Table 7: Pairwise correlation, mean and variance of response variables in inmates profiling data

	Correlation			Mean	Variance
	y_2	y_3	y_4		
y_1	0.0608	0.4993*	0.41421*	10.9373	85.4614
y_2		0.1200*	-0.0903*	0.7414	1.2458
y_3			0.2319*	4.6095	23.0024
y_4				2.2228	6.3477

* significant at 5% level ($n = 781$)

Both the MEEGR and MGPR models are fitted to the data set and the results are presented in Table 8. From the log-likelihood statistics, the MEEGR model performed better than the MGPR model. We carried out the test in equation (33) and we obtained the test statistic $Z_* = 10.46$ with $p\text{-value} < 0.0001$. Thus, the MEEGR outperform the MGPR in fitting the data. All the results about the dispersion parameters are the same for both regression models. They showed that all response variables are over-dispersed. Both models show that the number of arrests is positively affected by age and there is a significant difference between Black and White. The number of subsequent arrests is positively associated with the current offence severity in both models. Only the MEEGR model showed that female inmates have more subsequent arrests than male inmates. While age and Black positively influence number of LA custodies, the current offence severity negatively influences the number of LA custodies in both models. On the other hand, while there is a difference between Hispanic and White in the number of LA custodies under MEEGR model, there is no such difference under the MGPR model. The results from both models are about the same for the explanatory variables for the number of CDC custodies.

From Table 7, all bivariate correlations are significant at 5% level with the exception between y_1 and y_2 . All significant correlations are positive with the exception of correlation between variables y_2 and y_4 . Table 7 shows that all pairwise correlations are significant under the MEEGR model with the exception of correlation between variables y_2 and y_3 . Both models give only one significant negative correlation between y_2 and y_4 . Also, both models give the correlation between y_2 and y_3 to be insignificant.

Table 8: Parameter estimates (standard errors in parentheses) for inmate profiling data

	y_1		y_2	
Variable	MEEGR model	MGPR model	MEEGR model	MGPR model
Constant (x_0)	0.428 (0.254)*	0.910 (0.221)*	0.401 (0.364)	0.130 (0.343)
Age (x_1)	0.039 (0.004)*	0.029 (0.004)*	-0.010 (0.006)	-0.003 (0.006)
Rank (x_2)	0.0002 (0.003)	0.001 (0.003)	0.015 (0.004)*	0.013 (0.004)*
Gender (x_3)	0.159 (0.140)	0.175 (0.134)	-0.537 (0.213)*	-0.392 (0.206)
Black (x_4)	0.315 (0.096)*	0.428 (0.099)*	-0.132 (0.173)	-0.059 (0.165)
Hispanic (x_5)	0.095 (0.098)	0.150 (0.098)	-0.056 (0.172)	-0.109 (0.163)
$\hat{b}_t$	1.508 (0.103)*	0.178 (0.009)*	0.889 (0.091)*	0.373 (0.058)*

	y_3		y_4	
	MEEGR model	MGPR model	MEEGR model	MGPR model
Constant (x_0)	0.640 (0.168)*	0.555 (0.227)*	-0.769 (0.258)*	-0.528 (0.270)+
Age (x_1)	0.008 (0.003)*	0.014 (0.004)*	0.011 (0.004)*	0.014 (0.005)*
Rank (x_2)	-0.007 (0.003)*	-0.008 (0.003)*	0.001 (0.003)	0.002 (0.003)
Gender (x_3)	0.115 (0.085)	0.262 (0.143)	0.676 (0.175)*	0.740 (0.174)*
Black (x_4)	0.394 (0.081)*	0.551 (0.110)*	0.310 (0.115)*	0.280 (0.123)*
Hispanic (x_5)	0.297 (0.085)*	0.165 (0.109)	-0.013 (0.117)	-0.112 (0.120)
$\hat{b}_t$	1.214 (0.053)*	0.235 (0.014)*	1.439 (0.094)*	0.240 (0.021)*
$\hat{\rho}_{iv}$ for MEEGR	$\hat{\rho}_{12} = 0.1829 (0.0218)*$; $\hat{\rho}_{13} = 0.7146 (0.0406)*$; $\hat{\rho}_{14} = 0.2684 (0.0269)*$ $\hat{\rho}_{23} = 0.0227 (0.0436)$; $\hat{\rho}_{24} = -0.1049 (0.0215)*$; $\hat{\rho}_{34} = 0.2605 (0.0265)*$			
$\hat{\rho}_{iv}$ for MGPR	$\hat{\rho}_{12} = 0.0314 (0.0066)*$; $\hat{\rho}_{13} = 0.0177 (0.0105)$; $\hat{\rho}_{14} = 0.0380 (0.0130)*$ $\hat{\rho}_{23} = -0.0452 (0.0304)$; $\hat{\rho}_{24} = -0.1909 (0.0236)*$; $\hat{\rho}_{34} = 0.1402 (0.0294)*$			
Log-likelihood	For MEEGR: -6779.20		For MGPR: -6978.17	

* significant at 5% level

6 Summary and Conclusions

Famoye (2015) stated some advantages of the MGPR model when compared to the multivariate negative binomial regression model and the multivariate Poisson-lognormal regression model. Only the MGPR model is capable of modeling both over-dispersion and under-dispersion. The other two models are applicable to over-dispersed response count data. The MGPR model allows for correlations of any sign among count response variables and the correlations are independent of the dispersion parameters. Several multivariate models that can be used to fit several dependent response variables are proposed in this paper. Many of these multivariate models can be used for over-dispersed or under-dispersed count response variables. The models allow correlations of any sign.

In this paper, we define families of bivariate and multivariate generalized geometric regression models. These are based on the $T-R\{W\}$ framework. Many examples are given in the paper. The population means and variances of the response variables are not in closed form, but they are easy to compute because the CDF of the distributions are in closed form. It is easy to compute both the CDF and PMF for the distributions.

Two numerical data sets are used to illustrate the MEEGR model and it is compared with the MGPR model (See Famoye, 2015). In both data sets, the MEEGR model outperformed the MGPR model by using the goodness of fit test. The MEEGR model has closed form PMF and CDF while the MGPR model has closed form PMF and its mean and variance are in closed form. Both models can be used to fit data that exhibit over- or under-dispersion. The MGPR has a negative dispersion parameter for the under-dispersed situation while the dispersion parameter for the MEEGR model is always positive. Care must be taken when estimating the MGPR model for under-dispersed situation because the model gets truncated above. See Consul (1989, p. 4) or

Consul and Famoye (2006, p. 165) about the truncation error in the generalized Poisson distribution. This error is negligible when the number of nonzero probability classes is at least 5. In both numerical examples in Sect. 5, the response variable with the smallest range has seven non-zero probability classes.

In Sect. 4, we provide materials that can be used to compare multivariate count data regression models. This is illustrated in the application section (Sect. 5), where two generalized multivariate distributions and models are compared, and variables are selected using some of the test statistics described in Sect. 4. In this Chapter, a family of multivariate count data regression models is defined. Future work can include further inferences and a comparison between some of the models that were defined in the Chapter. New models can be defined and compared to the models in the Chapter.

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8 Chapter Appendix

In this Appendix, we will use the following notation: $\lfloor y \rfloor = \dot{y}$.

Table 2(i): Exponential-geometric{standard exponential}

By using Equations (6) and (8), we have

$$\begin{aligned} G(y) &= F_Y(y) = F_T\{\lambda Q_W(F_R(\dot{y}))\} = 1 - \exp\{\lambda \log(1 - F_R(\dot{y}))\} \\ &= 1 - (1 - F_R(\dot{y}))^\lambda = 1 - \varphi^{\lambda(\dot{y}+1)} = 1 - \theta^{\dot{y}+1}, \text{ where } 0 < \theta = \varphi^\lambda < 1. \end{aligned}$$

Table 2(iii): Exponentiated-exponential-geometric {standard exponential}

By using Equations (10) and (12), we have

$$\begin{aligned}
 G(y) = F_Y(y) &= F_T[Q_W(F_R(\dot{y}))] = \left[1 - \exp\{\lambda \log(1 - F(\dot{y}))\}\right]^\alpha \\
 &= \left[1 - \exp\{\log \phi^{\lambda(\dot{y}+1)}\}\right]^\alpha = \left[1 - (\exp[\lambda \log \phi])^{\dot{y}+1}\right]^\alpha \\
 &= \left[1 - \theta^{\dot{y}+1}\right]^\alpha, \text{ where } 0 < \theta = \exp[\lambda \log \phi] < 1, \alpha > 0.
 \end{aligned}$$

Table 2(iv): Exponentiated-exponential-geometric {standard log-logistic}

By using Equation (13), we have

$$\begin{aligned}
 G(y) = F_Y(y) &= F_T[Q_W(F_R(\dot{y}))] = \left[1 - \exp\{-\lambda F_R(\dot{y}) / [1 - F_R(\dot{y})]\}\right]^\alpha \\
 &= \left[1 - \exp\{-\lambda(1 - \phi^{\dot{y}+1}) / \phi^{\dot{y}+1}\}\right]^\alpha = \left[1 - \exp\{-\lambda(\theta^{-\dot{y}-1} - 1)\}\right]^\alpha, \\
 &\text{where } 0 < \theta = \phi < 1, \lambda > 0, \alpha > 0.
 \end{aligned}$$

Table 2(v): Weibull-geometric {standard exponential}

By using Equation (16), we have

$$\begin{aligned}
 G(y) = F_Y(y) &= F_T[\lambda Q_W(F_R(\dot{y}))] = 1 - \exp\{-\lambda F_R(\dot{y}) / [1 - F_R(\dot{y})]\} \\
 &= 1 - \exp\left\{-\left(\lambda Q_W(F_R(\dot{y}))\right)^c\right\} = 1 - \exp\left\{-\left(-\lambda \log[1 - F_R(\dot{y})]\right)^c\right\} \\
 &= 1 - \exp\left\{-\left(-\lambda \log \phi^{\dot{y}+1}\right)^c\right\} = 1 - \exp\left\{-\left(-\lambda \log \phi\right)^c(\dot{y}+1)^c\right\} \\
 &= 1 - \exp\left\{-\left(-\lambda \log \phi\right)^c(\dot{y}+1)^c\right\} = 1 - \theta^{(\dot{y}+1)^c}, \text{ where } 0 < \theta = e^{-(-\lambda \log \phi)^c} < 1, c > 0.
 \end{aligned}$$

Table 2(vii): Logistic-geometric {standard logistic}

By using Equation (18), we have

$$\begin{aligned}
 G(y) = F_Y(y) &= F_T[\lambda Q_W(F_R(\dot{y}))] = [1 + \exp\{-\lambda Q_W(F_R(\dot{y}))\}]^{-1} \\
 &= [1 + \exp\{\log(F_R(\dot{y}) / (1 - F_R(\dot{y}))^{-\lambda})\}]^{-1} = [1 + (F_R(\dot{y}) / (1 - F_R(\dot{y}))^{-\lambda})]^{-1} \\
 &= F_R^\lambda(\dot{y}) / [F_R^\lambda(\dot{y}) + (1 - F_R^\lambda(\dot{y}))^\lambda] \\
 &= (1 - \varphi^{\dot{y}+1})^\lambda / [(1 - \varphi^{\dot{y}+1})^\lambda + \varphi^{\lambda(\dot{y}+1)}], \text{ where } 0 < \theta = \varphi < 1, \lambda > 0.
 \end{aligned}$$

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